

Long-read sequencing of Nelore cattle using high molecular weight DNA from semen and Oxford Nanopore sequencing technology

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Introduction

Long-read sequencing is essential to investigate the prevalence of structural variants (SVs). The extraction of high molecular weight (HMW) genomic DNA, while challenging for many tissue and cell types, is a critical step for the success of this sequencing approach. Obtaining HMW DNA may be particularly challenging from mammalian sperm cells, where the DNA is highly compacted within a cell membrane rich in disulfide bonds. Despite these technical hurdles, frozen semen samples offer a strategic advantage for assessing population-level genetic diversity, enabling the inclusion of genetically influential animals that died decades ago.

Material and Methods

Twenty Nelore bulls were carefully selected based on, including progeny count and broad representation within the Brazilian herd. HMW genomic DNA was extracted with a modified version of the Gentra™ Puregene™ Tissue Kit (Qiagen) that was optimized for semen samples. DNA concentration and purity were assessed with a NanoDrop OneC spectrophotometer and the Qubit™ dsDNA BR Assay Kit. The DNA extractions yielded concentrations exceeding 300 ng/μL, with NanoDrop 260/280 ratios >1.8, and 260/230 ratios between 2.0 and 2.3, and final volumes around 50 μL, resulting in yields above 15,000 ng per sample (Figure 1). Average DNA fragment lengths exceeded 20 Kb. Libraries were prepared following the Ligation Sequencing Kit V14 (SQK-LSK114) protocol and sequenced on the Oxford Nanopore Technologies (ONT) PromethION platform (R10.4.1 pores) in 72-hour runs. For each sample, ~200 ng was loaded onto a FLO-PRO114M flow cell, followed by a mid-run wash (after ~25 hours) and reloaded of an additional 200 ng (Figure 2.A). Real-time basecalling was performed using the fast accuracy model, and reads were subsequently rebasecalled using the super accuracy model (SUP) in Dorado. Adapter and barcode sequences were trimmed using Porechop. Subsequently, reads were filtered with SeqKit2 using the default ONT quality threshold (>Q10 for the SUP model). Reads were aligned to the bovine reference genome ARS-UCD2.0 using Minimap2. The resulting BAM files were sorted and indexed with Samtools.

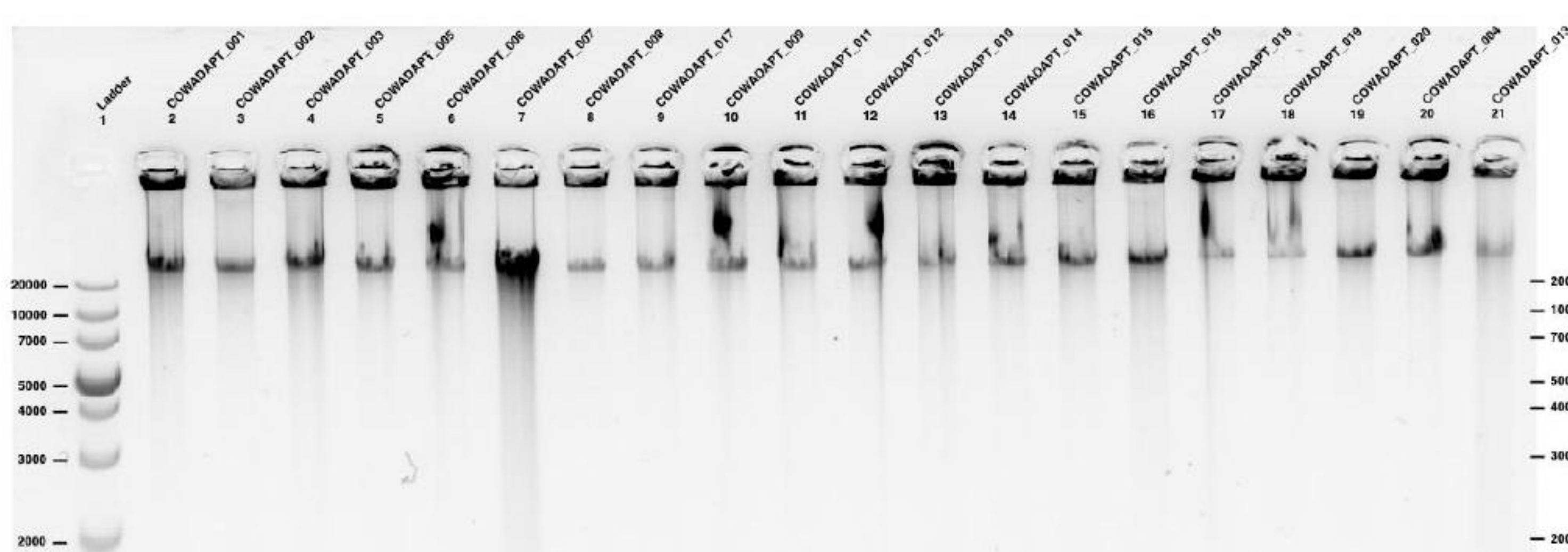


Figure 1. Agarose gel electrophoresis (1%) of HMW DNA isolated from bovine semen samples.

Results and Discussion

Table 1. Overview of fast basecalled data for the 20 bovine samples.

Sample	Basecalling model	Estimated bases (Gb)	Coverage	Yield fast pass (Gb)	Yield fast fail (Gb)	Estimated read N50 (Kb)
COWADAPT_001	doradov0.6.1_SUP	65,96	24,43	40,20	22,90	9,70
COWADAPT_002	doradov0.7.2_SUP	46,55	17,24	31,30	13,90	11,60
COWADAPT_003	doradov0.7.2_SUP	67,29	24,92	59,30	6,80	9,20
COWADAPT_004	doradov0.7.2_SUP	42,95	15,91	24,20	16,45	9,40
COWADAPT_005	doradov0.7.2_SUP	84,81	31,41	76,30	8,40	8,40
COWADAPT_006	doradov0.7.2_SUP	46,95	17,39	42,10	4,40	10,30
COWADAPT_008	doradov0.7.2_SUP	45,52	16,86	36,59	8,76	12,69
COWADAPT_009	doradov0.7.2_SUP	73,35	27,17	65,80	6,90	11,10
COWADAPT_010	doradov0.7.2_SUP	84,75	31,39	77,90	8,00	9,30
COWADAPT_011	doradov0.7.2_SUP	82,73	30,64	74,30	8,70	11,60
COWADAPT_012	doradov0.6.1_SUP	59,59	22,07	45,30	11,80	9,80
COWADAPT_013	doradov0.7.2_SUP	63,96	23,69	54,50	9,70	11,30
COWADAPT_014	doradov0.7.2_SUP	57,07	21,14	52,30	5,00	13,70
COWADAPT_015	doradov0.7.2_SUP	73,77	27,32	43,90	26,40	11,00
COWADAPT_016	doradov0.7.2_SUP	67,62	25,04	50,80	6,00	9,40
COWADAPT_017	doradov0.7.2_SUP	98,51	36,49	85,08	13,07	10,53
COWADAPT_018	doradov0.7.2_SUP	64,11	23,74	36,00	24,90	9,10
COWADAPT_019	doradov0.7.2_SUP	121,30	44,93	102,90	18,53	8,40
COWADAPT_020	doradov0.7.2_SUP	100,05	37,06	85,20	14,10	10,10
COWADAPT_023	doradov0.7.2_SUP	72,25	26,76	59,88	11,41	12,55

Each sample produced over 40 Gb of raw sequencing data (Table 1), achieving ~15× genome coverage. Following quality filtering (>Q10), more than 31.4 Gb of high-quality reads per sample were retained, with an average N50 read length of 10.44 Kb (Figures 2.B and 2.C).

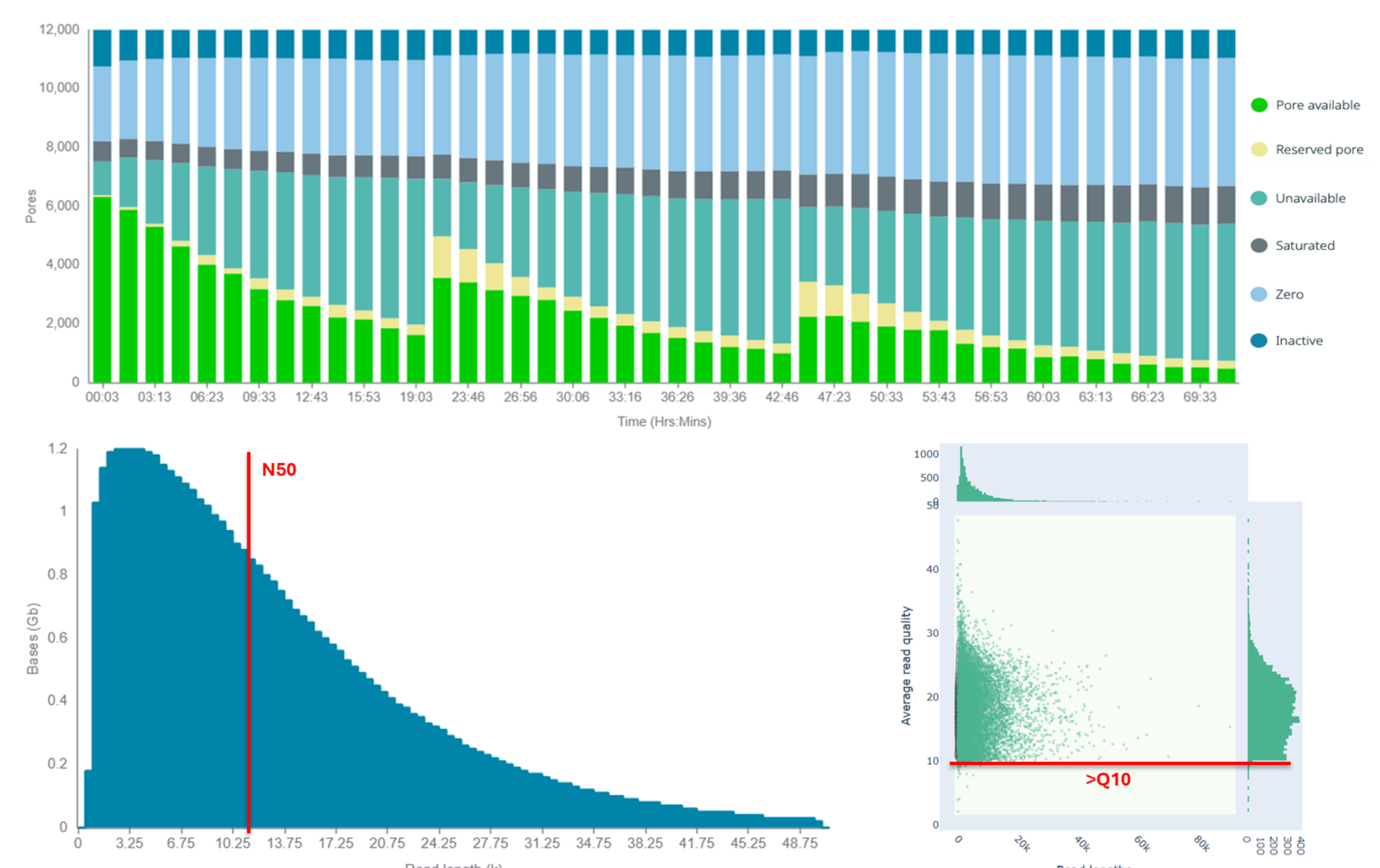


Figure 2. A) Scan to determine pore status in Flowcell channels. B) Total number of bases relative to read length. C) Read length vs. average read quality.

Conclusions

These results confirm that, despite the intrinsic challenges of working with semen-derived DNA in long-read sequencing workflows, the extractions produced sufficient amounts of data that will facilitate the detection and characterization of common SVs.

Acknowledgements